

ASPN Platform Developer Handoff Guide

September MVP Foundation (June 2026)

Project Overview

The Aspirations Network (ASPN) Platform is a youth development, credentialing, civic engagement, workforce readiness, and opportunity platform.

Primary user population:

- K–12 students
- College students
- Emerging adults

Target age range:

None

8–24 years old

The platform is intended to function as:

None

Youth Portfolio

+

Credential Wallet

+

Program Participation System

+

Civic Opportunity Network

The platform is NOT intended to function as:

None

Social Media

Messaging Platform

Follower Network

Public Youth Directory

Youth safety, privacy, and staff oversight take priority over engagement features.

Current Repository

Repository:

None

AspirationsBackendColby

Backend project:

None

UserData

Technology stack:

None

Java 21

Spring Boot

Gradle

Firebase Admin SDK

Firestore

Hosting direction:

None

Firebase

Current status:

None

September MVP Backend Foundation Complete
Firebase Staging Validation Pending
Frontend Integration Pending

Development Philosophy

When making development decisions ask:

None

Does this help youth safely document accomplishments,
earn credentials,
participate in ASPN programs,
and access future opportunities?

If not, reconsider the feature.

Features Explicitly Out of Scope

Do NOT build without explicit approval:

- Direct messaging
- Peer-to-peer messaging
- Public youth profile discovery
- Follower systems
- Friend systems
- Like-driven popularity systems
- Infinite scrolling
- Viral recommendation systems
- Public youth search

Platform safety is prioritized over engagement.

Current Backend Architecture

Current backend supports:

Profiles

Youth profile creation

Youth profile retrieval

Private profiles by default

Profile status tracking

Program participation tracking

Staff review flags

Staff verification flags

Credentials

Credential definitions

Earned credentials

Staff-created credentials

Staff-awarded credentials

Credential retrieval for display

Credential association with youth profiles

Attendance

Attendance record model

Attendance creation

Attendance retrieval

Attendance association with youth profiles

Supported statuses:

None

present

absent

excused

pending

Service Hours

Service-hour record model

Service-hour creation

Service-hour retrieval

Service-hour association with youth profiles

Supported statuses:

None

pending

verified

rejected

Placeholder exists for future Google Form integration.

Google Form integration has NOT been built.

Staff User Management

Staff can:

- List youth users
- Retrieve youth users
- Update onboarding fields

Staff cannot currently:

- Change roles
- Create public profiles
- Manage permissions beyond existing workflow

Authentication

Firebase ID token verification implemented.

Protected staff routes require:

None

`Authorization: Bearer <Firebase ID token>`

Role check required:

None

`staff`

`or`

`admin`

Current Firestore Collections

Existing collections:

None

`aspirationnetworkusers`

`aspirationnetworkposts`

`comments`

`credentialDefinitions`

```
earnedCredentials
attendanceRecords
serviceHourRecords
```

Important User Fields

Current user documents should support:

```
None
uid
role
youthProfile
publicProfile
profileStatus

programIds
programParticipationIds

earnedCredentialIds
attendanceRecordIds
serviceHourRecordIds

staffReviewRequired
staffVerified
```

Current Core User Flow

Youth signs in through Firebase

↓

Backend validates Firebase token

↓

Youth retrieves:

None

```
GET /api/me/profile
```

↓

Backend returns:

- profile information
- credentials
- attendance records
- service-hour records

Youth cannot retrieve another youth profile.

Current Core Staff Flow

Staff signs in

↓

Backend validates Firebase token

↓

Backend verifies:

None

```
role = staff  
or  
role = admin
```

↓

Staff can:

- manage youth users

- create credential definitions
 - award credentials
 - create attendance records
 - create service-hour records
-

Major Checkpoints Completed

Checkpoint 1

Backend cleanup

- collection mismatch fixed
 - deletion cleanup fixed
 - error handling improved
-

Checkpoint 2

Profile Foundation

Added:

- onboarding status
 - participation fields
 - attendance fields
 - service-hour fields
-

Checkpoint 3

Credential Foundation

Added:

None

CredentialDefinition

EarnedCredential

Checkpoint 4

Staff Credential Management

Added:

None

POST /api/staff/credentials/definitions

POST /api/staff/credentials/award

Checkpoint 5

Credential Display

Added:

credential retrieval for profile display.

Checkpoint 6

Authentication & Role Protection

Added:

Firebase token verification

Staff/Admin role protection

Checkpoint 7

Staff User Management

Added:

None

GET /api/staff/users/youth

GET /api/staff/users/youth/{id}

PATCH /api/staff/users/youth/{id}

Checkpoint 8

Attendance Foundation

Added:

None

POST /api/staff/attendance

GET /api/staff/attendance/user/{userID}

Checkpoint 9

Service-Hour Foundation

Added:

None

POST /api/staff/service-hours

GET /api/staff/service-hours/user/{userID}

Checkpoint 10

Youth Self-Service Profile

Added:

None

GET /api/me/profile

Checkpoint 11

Firebase Staging Readiness Review

Backend determined ready for staging validation.

Current State Assessment

Backend is:

None

Structurally Ready

for:

- profile-based onboarding
- credential assignment
- attendance tracking
- service-hour tracking

for approximately:

None

100 youth users

during initial pilot deployment.

Immediate Development Priorities

Priority 1:

Firebase staging deployment

Priority 2:

Frontend integration

Priority 3:

Real user testing

Priority 4:

Firestore rules review

Recommended Next Development Phase

Phase II: Frontend Integration

Build:

Youth Profile Page

Display:

- profile information
- credentials
- attendance
- service hours

Uses:

None

GET /api/me/profile

Staff Dashboard

Support:

- youth user management
 - credential creation
 - credential awarding
 - attendance management
 - service-hour management
-

Credential Catalog Management

Allow staff to:

- create credentials
- edit credentials
- deactivate credentials

Credential names are intentionally NOT hardcoded.

Future Platform Roadmap

Potential future phases:

Program Management

Programs

Cohorts

Projects

Participation records

Workforce Pathways

Government pathways

Internship pathways

Career pathways

Scholarship pathways

Civic Engagement Features

Safe discussion boards

Moderation tools

Audit logging

Structured civic participation

Public Visibility (Future Discussion)

Any future public-profile system should require:

- safety review
- privacy review
- staff approval process
- age considerations

No public youth discovery currently exists.

Development Rule

Preserve:

None

Spring Boot

Firebase

Firestore

architecture unless there is a compelling reason to change.

Favor:

None

Maintainability

Auditability

Security

Youth Safety

over rapid feature expansion.

The mission of the platform is more important than feature quantity.

END OF DEVELOPER HANDOFF GUIDE